Robust Federated Learning for Privacy Preservation and Efficiency in Edge Computing

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***Abstract*—Federated Learning (FL) has emerged as a key enabler of privacy-preserving distributed model training in edge computing environments, crucial for service-oriented applications such as personalized healthcare, smart cities, and intelligent assistants. However, existing privacy-preserving FL methods are susceptible to multiple privacy leakage attacks (MPLA), where adversaries infer sensitive information through repeated gradient updates. This paper proposes a Robust and Communication-Efficient Federated Learning (RCFL) framework designed to enhance privacy protection and communication efficiency in edge-based service environments. RCFL integrates a global privacy-preserving mechanism with an innovative privacy encoding strategy that minimizes privacy risks over multiple data releases while significantly reducing communication overhead. The proposed framework's theoretical analysis demonstrates its ability to maintain differential privacy across numerous interactions, ensuring robust model convergence and efficiency. Experimental results using MNIST and CIFAR-10 datasets reveal that RCFL can lower the MPLA success rate from 88.56% to 42.57% compared to state-of-the-art methods, while reducing communication costs by over 90%. These findings underscore RCFL's potential to enhance security, efficiency, and scalability in service-oriented edge computing applications.**

***Index Terms*—Federated learning, edge computing, multiple privacy leakage attack, privacy-preserving.**

I. Introduction

With the development of Big Data and the Internet of Things, machine learning, especially deep learning, is increasingly applied in various FL scenarios. Applications include facial recognition [20], [24], [14], medical diagnosis [25], [16], [38], and interactive intelligent assistants [29], [41], [21], to provide better services while protecting sensitive information. Although distributed clients in FL only send model updates, such as updated gradients or model parameters, and do not upload raw datasets [4], the gradients and model parameters uploaded by clients still pose risks of privacy leakage [18]. For example, attackers can infer private information from gradients uploaded by clients [19], [22]. Furthermore, adversaries can deduce the types of data held by clients participating in training and reconstruct the original data from the updated model information [9] [33]. With the evolution of model inversion attacks and gradient inversion attacks, attackers can achieve high accuracy in recovering private information, such as personal details, from model data, posing a significant security risk for participants in FL [15].

Recent model inversion attacks focus on the correlation between model inputs and outputs in published models. These attacks reconstruct images by inferring input data from output categories [32], [26], [40]. Directly weakening the dependency between inputs and outputs is an effective way to defend against model inversion attacks [32]. Specifically, [32] introduces regularization functions in models to penalize mutual information between inputs and outputs, thereby reducing their correlation. Reducing the correlation (or mutual information) between inputs and outputs is a common strategy to defend against model inversion attacks. However, this approach can significantly degrade the model's utility. When the correlation falls below a certain threshold, the model can no longer learn from the input data. Additionally, calculating mutual information requires substantial computational resources, which is a significant challenge for resource-limited edge devices. To reduce input-output correlation while improving model utility, [26] add dependency constraints in model training, reducing redundant information and increasing distinct representations in posterior layers, thereby enhancing model accuracy while defending against Model Inversion Attacks (MIA). Furthermore, introducing noise directly into model outputs can also confuse the relationship between model outputs and inputs [40]. Specifically, [40] adds differential privacy noise at both the category and subcategory levels to protect client privacy. However, this approach still compromises between privacy protection and model utility. Existing solutions are tailored for centralized or single-release privacy protection, but data owned by participants in FL and continually generated for training is often a series of highly correlated data (e.g., a series of selfies and specific pet photos, or different photos of the same target). To obtain better personalized services, participants in FL need to continually train on newly generated, highly correlated private data. This continuous release of model updates

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in the same category or with the same features makes participant privacy more prone to leakage. Therefore, privacy protection methods for single data releases may be insufficient to prevent multiple privacy leakage attacks.

In this paper, we investigate MPLA against FL for edge computing. We demonstrate that existing defense methods are ineffective against MPLA (a single model inversion attack success rate of 31%, increasing to 82.12% after ten releases, and 88.56% after twenty releases). To defend MPLA, we propose a robust and communication-efficient FL framework (RCFL) for edge computing. Specifically, we propose a global privacy-preserving mechanism and privacy encoding mechanism to ensure privacy during continuous data releases of edge clients and communication efficiency.

Experiments show that RCFL demonstrates state-of-the-art defense performance across multiple datasets, target models, and against continuous privacy leakage attacks. Notably, to defend against attacks, the classification accuracy of MID [32], BiDO [26], and class-DP [40] all drop by over 5% compared to undefended classifiers. Also, the communication overhead of RCFL is less than other solutions. These experimental results show the effectiveness of RCFL in edge computing and inspire a new direction for defending against MPLA attacks. Our main contributions are summarized as follows:

- We investigate Multiple Privacy Leakage Attacks for FL in edge computing scenarios, and demonstrate that existing defense methods are ineffective against this attack (attack success rate exceeds 88.56%).
- Against MPLA, we propose a robust and communication-efficient federated learning framework for edge computing to ensure security and efficiency.
- RCFL proposes a global privacy protection mechanism to reduce the attack accuracy of MPLA from 88.56% to 42.57%.
- RCFL proposes a privacy gradient encoding mechanism to reduce communication overhead from 381.5 MB to 11.9 MB.
- We provide theoretical and empirical analysis of the security and efficiency of RCFL and RCFL's robustness and communication efficiency for edge computing.

The structure of this paper is as follows: Section II reviews the preliminaries. Section III introduces the proposed Multiple Privacy Leakage Attacks. Section IV details the proposed RCFL. Section V analyzes the security and efficiency of RCFL. Section VI provides experimental results obtained using RCFL. Section VII analyzes the related work. Finally, Section VIII concludes the paper and highlights potential areas for future work. A summary of basic notations is provided in Table I.

TABLE I
NOTATIONS

Symbol	Meaning
n	number of clients
E_i	i-th edge client
p_i	intermediate parameters of E_i
F_i	loss function of E_i
GF	global loss function
g_i	the gradient of E_i
g_i^*	the perturbed gradient of E_i
α	online rate of edge clients

II. Preliminaries

This section introduces differential privacy, global privacy of FL, and the threat model.

A. Differential Privacy

Differential privacy maximizes the accuracy of queries from statistical databases while minimizing the risk of identifying individual entries. In FL, it ensures the privacy of individual data points during collective learning.

Definition 1. ϵ-differential privacy: A randomized function K is ϵ-differentially private if for any two datasets D_1 and D_2 differing by one element, and all subsets S of K's range,

$$Pr[K(D_1) \in S] \leq e^{\epsilon} \times Pr[K(D_2) \in S],$$

where ϵ is a small positive number [6].

Another important concept in differential privacy for determining the noise scale in FL for edge computing is the sensitivity.

Definition 2. Sensitivity: For function $f : D \to R^n$, sensitivity Δf is defined as:

$$\Delta f = \max_{D_1, D_2} ||f(D_1) - f(D_2)||_1,$$

where D_1 and D_2 are datasets differing by one element and $\Pr[\cdot]$ denotes probability [6].

Definition 3. (ε, δ)-Differential Privacy: A randomized mechanism $\mathcal{M}$ satisfies (ε, δ)-differential privacy if for any two adjacent datasets D and D' (datasets that differ by only one element), and for all S in the output range of $\mathcal{M}$, the following inequality holds:

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(D') \in S] + \delta$$

Where: ε (epsilon) is a non-negative parameter that provides an upper bound on how much the probability of any output can increase by changing a single element in the dataset. A smaller ε means better privacy, δ is a small positive parameter, which represents the probability of the mechanism $\mathcal{M}$ failing to give ε-differential privacy.

The noise added to the gradient in FL for edge computing generally uses Gaussian noise.

Lemma 1. Gaussian Mechanism: Given a function f with sensitivity Δf, $f'(x)$ satisfies (ϵ, δ)-differential privacy if:

$$f'(x) = f(x) + N(0, \sigma^2),$$

with $N(0, \sigma^2)$ being the Gaussian distribution with mean 0 and standard deviation σ, and $\sigma \geq \Delta f \frac{\sqrt{2 \ln (1.25/\delta)}}{\epsilon}$ [6].

Another important concept in differential privacy relevant to ensuring privacy in FL for edge computing is the Post-Processing Property of Differential Privacy.

Definition 4. Post-Processing Properties of Differential Privacy: If a mechanism K is (ε, δ)-differentially private, and g is a function, then $g \circ K$ also preserves (ε, δ)-differential privacy [6].

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B. Global Privacy of FL for Edge Computing

We explore the delicate balance between privacy preservation and the iterative nature of FL for edge computing. Our focus zeroes in on the interrelation between the frequency of model updates in FL for edge computing, where edge clients utilize local data for model training, and the consequent escalation in privacy breach risks. This issue is particularly acute in edge-computing scenarios involving sensitive personal data.

Our methodology involves a two-step process executed sequentially by the Central Server and each edge client:

- *Security Level Determination:* This includes monitoring the online status and privacy budget usage of the CS's edge clients and selecting participants for the current training round.
- *Data Privacy Protection for Multiple Releases:* Edge clients introduce adaptive noise based on the times of gradient updates and submit the encoded gradient.

We recognize the challenges posed by adversaries who might attempt to infer client privacy by intercepting communication channels. We analyze and demonstrate that multiple data releases from edge clients enable the CS to decode most local models in a reasonable time frame for the state-of-art schemes [32], [26], [40]. Our analysis not only underscores the inherent privacy challenges in FL for edge computing but also demonstrates the need for global privacy of FL for edge computing to assess and mitigate these risks.

C. Threat Model

The threat model in FL for edge computing provides an in-depth analysis of the vulnerabilities inherent in the interactions between edge clients and the central server. The model posits that ECs, while generally dependable in model updates and uploads, might inadvertently expose sensitive data, highlighting a key area of vulnerability. This exposure becomes particularly pertinent given the semi-honest nature of the CS, which, despite performing its functions accurately, may have underlying interests in accessing ECs' private data. Specifically, the central server may utilize the relevance of client data published multiple times $\{w_i^1, w_i^2, ..w_i^T\}$ of edge client E_i to reconstruct the original client data and infer its labels. This dual nature of the CS's role presents complex security challenges, particularly its potential to infer private information from gradients submitted by ECs.

III. Multiple Privacy Leakage Attack

According to the system model and threat model, we notice that the client's multiple releases of a set of highly relevant data may increase the risk of privacy leaks. In this section, we investigate MPLA and demonstrate that existing privacy-preserving FL schemes are ineffective against this attack (attack success rate exceeds 88.56%).

MPLA targets multiple releases in FL for edge clients and is aimed at bypassing classic defense algorithms designed to resist model inversion and gradient inversion attacks, including MID [32], BiDO [26], and class-DP [40]. We detail MPLA in terms of proposed MPLA, implementation configuration, attack process, and attack evaluation.

Algorithm 1: MPLA: Multiple Privacy Leakage Attack.

Require: Aggregation parameters $\{w_1, w_2, .., w_T\}$ and gradients list $\{g_i^1, g_i^2, .., g_i^T\}$ of the targeted client E_i.

```
for t = 1; t ≤ T; t++ do
  for j = 1; j ≤ N; j++ do
    Calculate dummy gradients as Equ. (3)
    Calculate the distance as Equ. (2)
    Update the sample set Sa_i^t according to Equation (4).
    Update: Label'_i as Equ. (5)
  end for
end for
return {label'_i^1, label'_i^2, ..., label'_i^T} and
  {Sa_i^1, Sa_i^2, ..., Sa_i^T}.
```

A. Proposed MPLA

Multiple privacy leakage attacks exploit two characteristics of the interactions between the central server and edge clients:

1) The private information stored on edge clients is highly correlated, and the data within a certain period belong to the same category (such as a series of selfies, avatars, and pets). Due to their quantity and category, their data distribution is non-independent and identically distributed.
2) The potential features contained in the model gradients received by the central server from the same edge client have consistency. As gradients are continuously submitted, the accuracy of privacy leakage attacks (such as model inversion attacks, and membership inference attacks) by the central server will continuously improve.

We hypothesize that as updates are submitted, edge clients become more susceptible to privacy leakage attacks by semi-honest central servers as updates are submitted. Preventing the leakage of private information of edge clients from multiple privacy leakage attacks by the central server is the cornerstone of ensuring privacy and security in FL for edge computing [13], [12], [37]. Typically, the CS can obtain the client's private information through multiple inferences.

A multiple leakage attack aims to exploit multiple data points or features simultaneously to reconstruct sample data from a model's output. This approach is particularly effective when samples share characteristics or belong to similar categories.

The MPLA algorithm is as shown in Algorithm 1. We assume that the edge client E_i possesses a set of data D_i with the same label $label_i$ (such as a series of selfies and a series of pet photos) [22], [19], $D_i = \{D_{i,1}, D_{i,2}, \ldots, D_{i,m}\}$, where m is the number of samples. E_i trains the local model on the partitioned data and uploads the generated gradient g_i. The purpose of MPLA is to infer the client's original data and data labels by fitting the gradients uploaded by the edge clients multiple times,

$$label_i'^{*} = Top_k\{label_i'^{1}, label_i'^{2}, \ldots, label_i'^{t}\}, \quad (1)$$

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where Top_k represents the top k categories, and $label_i'^t$ is the label fitted by the adversary based on g_i^t. The execution process of MPLA is as follows:

The adversary's acquisition of the target's original data and labels can be formalized as:

$$Sa_i^{t+1}, label_i'^{t+1} = \arg \min_{Sa_i^t, label_i'^t} ||g_i^t - g_i'^t||^2, \quad (2)$$

where Sa_i and $Label_i'$ are the virtual samples and virtual labels generated locally by the attacker, g_i^t is the updated gradient of the targeted client E_i, and

$$g_i'^t = \nabla F_i(w_t, Sa_i^t), \quad (3)$$

where F_i is the local model's loss function, Sa_i^t is the dummy sample for edge client E_i, and w_t is the model weight at the t-th iteration.

We use the gradient descent method to optimize the process of the virtual sample and virtual label Sa_i and $Label_i'$ as follows:

$$Sa_i^{t+1} \leftarrow Sa_i^t - \beta \nabla_{Sa_i^t} ||g_i^t - g_i'^t||^2, \quad (4)$$

$$label_i'^{t+1} \leftarrow label_i'^t - \beta \nabla_{label_i'^t} ||g_i^t - g_i'^t||^2, \quad (5)$$

where β is the learning rate. We get the dummy $g_i'^t$ by feeding Sa_i^{t+1} and $label_i'^{t+1}$ and iteratively update Sa_i^{t+1} and $label_i'^{t+1}$ until the distance between $g_i'^t$ and target g_i^t is close enough. For each g_i^t, we updates $g_i'^t$ for N times.

To illustrate the label update process, we take softmax as the last layer of the neural network as an example. The softmax layer converts logits into probabilities. Gradients are computed concerning these logits. This is done by differentiating the loss function concerning each logit. In a privacy attack, the central server would have gradients from edge clients in federated learning, and it'd use these to reconstruct the logits. Apply gradient descent to update the logits as Equ. (2). After updating the logits, apply the softmax function again to get the updated probabilities. These probabilities reflect the network's confidence in each class after the gradient descent step. Once updated probabilities, the central server can derive the class labels by selecting the class with the highest probability (argmax).

At t-th iterations of MPLA, the adversary obtains the label $(label_i'^{t+1}, Sa_i^{t+1})$ corresponding to $(g_i^t, Sa_i^t, label_i'^t)$. After T iterations, MPLA obtains $\{label_i'^1, label_i'^2, \ldots, label_i'^T\}$ and infers the target label through Top_k.

B. Implementation Configuration of MPLA

To validate our hypotheses, we establish an edge computing environment. Training sessions are carried out on two datasets: the standard MNIST dataset[1] and the CIFAR-10 dataset.[2]

In our setup, edge clients are equally likely to be online and partake in the training process. This uniform distribution ensures that each client has an equal probability of contributing to each training round, providing a balanced representation of client data. Each client is assigned a distinct number of data samples, ranging from 1 to 20. This variation in sample size among clients introduces a degree of non-IID data distribution, crucial for testing the robustness of the federated learning models against privacy attacks. To manage the gradient updates from clients, we employ a Poisson distribution mechanism at the central server. The server only accepts a specific number of gradient updates for each training round. The acceptance rate is governed by a predefined lambda parameter, aligning with the principles of a Poisson distribution. Gradients that exceed the threshold set by the Poisson distribution are not considered for that particular training iteration. This decision is crucial for simulating a realistic edge computing environment where network bandwidth and server processing capabilities are often limited.

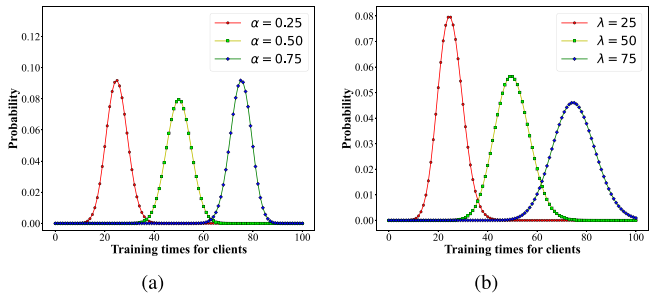


Fig. 1. (a) and (b) compare the probability of different training times for edge clients with the Uniform distribution and the Poisson distribution, respectively. (a) shows the probability of different training times with Uniform distributions; (b) shows the probability of different training times for edge clients with Poisson distributions.

The number of training times is $T = 100$, and the number of edge clients is $n = 200$, respectively. To demonstrate the performance of MPLA on Non-IID, we assume online edge clients have different training iterations and obey Uniform distributions and Poisson distributions respectively. Online rates are α and λ for Uniform distributions and Poisson distributions. A single attack uses a setting of [5] for MID [32], BiDO [26], and class-DP [40]. Given the unstable connections between edge clients and the CS, the online distribution of edge clients is presumed to adhere to a Poisson distribution [10], [27] or a uniform distribution [3], [34]. The attack strategy is implemented on a server running Ubuntu 20.04 LTS. The central server functions on a system equipped with an Intel Core i7-8700 processor with 32 GB system RAM.

C. MPLA Process

Depending on different client distributions, the number of times clients participate in training significantly increases with the online probability. Fig. 1 details the relationship between the number of times clients participate in training and the corresponding probabilities. Specifically, as shown in Fig. 1, when the client online rate is $\alpha = 0.75$ or $\lambda = 50$, after 100 iterations, more than half of the clients participate in training more than 50 times. Conversely, when the online rate is lower, such as $\alpha = 0.25$ or $\lambda = 25$, more than half of the clients participate in training less than 30 times. Therefore, with the increase in online rates, changes in client distributions, and the frequency

[1] http://yann.lecun.com/exdb/mnist/
[2] https://www.cs.utoronto.ca/kriz/cifar.html

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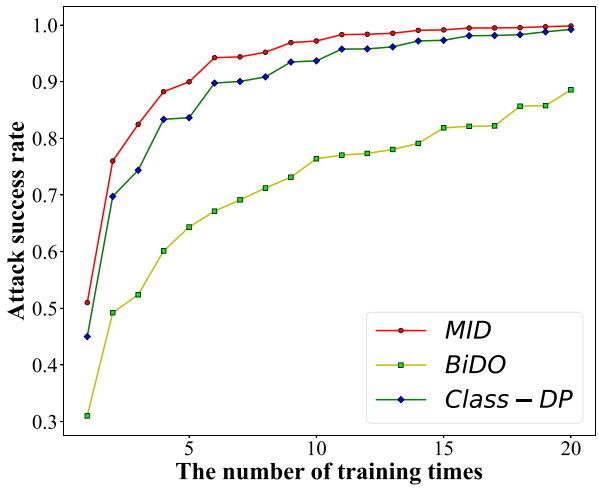


Fig. 2. Attack success rate for different algorithms.

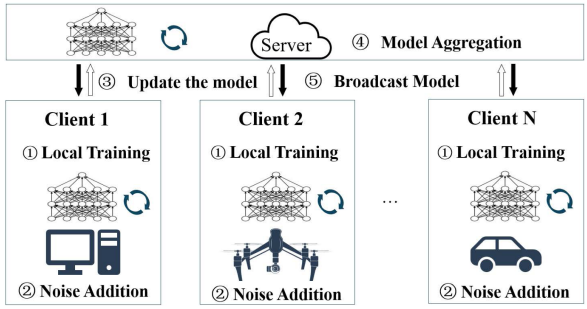


Fig. 3. The system model of RCFL.

of model updates, the number of times clients participate in training significantly increases.

We set up different client distributions and online scenarios for the experiment. Clients train a photo each time, and the samples trained belong to the same person (the same target or the same number). After receiving gradients uploaded by a specific client, the server attempts to recover its original data. As the number of gradients uploaded by the target client increases, the server performs clustering and inference on the recovered data. If the majority of the reconstructed data belongs to the same category, we reconstruct the original data through data augmentation and infer that the original data belongs to this category. We set various numbers of client participation in training (1-100) to observe the corresponding attack success rates for MID [32], BiDO [26], and class-DP [40] on CIFAR-10.

D. Evaluation of MPLA

To measure the ability of MPLA, we classify the samples after $T = 100$ reconstructions. If the classifier achieves a certain level of accuracy, the attack is considered to be successful. Fig. 2 summarizes the MPLA attack results on MID [32], BiDO [26], and class-DP [40]. In our analysis of attack outcomes within FL for edge computing context, a critical observation emerged regarding the success ratio and efficiency of multiple privacy leakage attacks. Empirical evidence demonstrates a substantial correlation between the frequency of client participation in the training process and the consequent risk of privacy breaches. Specifically, it was observed that with an increase in the number of training participations by a client, the likelihood of successful information extraction escalates markedly. This phenomenon is particularly pronounced when client engagement exceeds ten training sessions, at which point the success rate of privacy invasion consistently surpasses the threshold of 95%. Such a high success rate not only reflects the acute vulnerability inherent in the FL systems for edge computing but also underscores the efficiency of the attack strategy. The relative ease with which private data can be compromised, even with limited client-system interactions, poses a significant challenge to the security protocols employed in FL for edge computing networks. This vulnerability is particularly concerning in scenarios characterized by frequent data contributions from the same set of clients. The findings of our study necessitate a re-evaluation of current security measures in FL for edge computing environments, advocating for the integration of more robust, privacy-preserving methodologies to safeguard against the escalating risks of data privacy infringement. Additionally, the non-IID personal datasets cause slower model convergence and overfitting issues. The prediction accuracy is 97.5% for IID datasets and 95.39% for non-IID datasets.

IV. Construction of RCFL

To defend MPLA, we propose a Robust and Communication-Efficient Federated Learning (RCFL) framework for edge computing. RCFL ensures global privacy throughout multiple gradient updates. It also reduces the communication overhead, which is crucial for resource-constrained edge clients. This section describes the RCFL framework and details the main technologies of RCFL. The details of RCFL are provided in Algorithm 2.

A. Overview on RCFL

RCFL addresses the challenges of privacy against MPLA for edge computing scenarios. It involves a central server and multiple edge clients, with RCFL's operation divided into two main stages: multiple privacy budget allocation optimization, and privacy encoding and global privacy aggregation. This investigation aims to achieve a balance between maintaining client privacy and minimizing communication overhead in federated learning for edge computing. The overview of RCFL are detailed in Fig. 3.

In RCFL, edge clients collaborate with the central server to train a globally optimal model while ensuring privacy. A key aspect of this approach is adding Gaussian noise to each client's model gradient, which adheres to differential privacy standards. RCFL controls the total privacy budget for each client across multiple training sessions, using allocation methods like geometric or P-series to cater to the times of client participation due to fluctuating network conditions.

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Algorithm 2: RCFL: Robust and Communication-Efficient Federated Learning Framework for Edge Computing.

Require: CS broadcasts w^0 to all ECs.
for $t = 1; t \leq T; t++$ **do**
  while Edge clients $\in \{E_i\}$ **do**
    E_i locally trains the model as Equ. (8)
    E_i allocates the privacy budget as Equ. (12)
    Perturb the gradients as Equ. (9)
    E_i encodes the gradients as Equ. (17)
    E_i sends the updates to CS
  end while
  CS aggregates gradients as Equ. (18)
  CS broadcasts w^t to all online ECs
end for
return w^T.

RCFL also introduces a privacy encoding mechanism, where each client encodes their noise-added gradient to mask privacy leaks, leveling the gradient and thus eliminating private features. The central server aggregates these gradients, add noise to aggregation results based on the privacy budget, and then broadcasts the updated parameters. This step-by-step algorithmic approach ensures that RCFL is robust and efficient for federated learning in edge computing environments, effectively balancing the need for privacy and communication efficiency.

B. The Proposed Scheme

The primary objective of RCFL is twofold: ensuring global privacy security in federated learning for edge computing, and reducing the communication overhead for edge clients. The specific implementation process of RCFL is shown in Algorithm 2. Specifically, RCFL includes two stages: global privacy protection, and gradient privacy encoding and aggregation.

The goal of the RCFL framework is to collaboratively train a global model with the participation of multiple edge clients while ensuring privacy at each step. The training process can be represented as:

$$w = \arg\min_{w} GF(w, D) \tag{6}$$

where w represents the global model parameters and $GF(w, D)$ is the global loss function that depends on the model parameters w and the dataset D, which is the collection of datasets owned by all clients. This loss function is minimized by optimizing the global model across all iterations. Each edge client E_i computes its local model update based on its private dataset D_i. The local update is computed using gradient descent as:

$$w_i^{t+1} = w_i^t - \beta \nabla F_i(w^t, D_i) + \frac{\beta}{2} \nabla F_i(w^t, D_i)^2 \tag{7}$$

where w_i^t is the local model parameters of client E_i at iteration t, β is the learning rate for client E_i, and $\nabla F_i(w^t, D_i)$ is the gradient of the local loss function F_i with respect to the global model parameters w^t. The second term $\frac{\beta}{2} \nabla F_i(w^t, D_i)^2$ is the regularization term to reduce overfitting. Each client computes and uploads its updated model, and the central server aggregates the updates to obtain the new global model. The aggregation process ensures that the global model is refined based on the contributions of each client.

Single Privacy Protection: After receiving the initial model parameters w^0 from the server, the i-th client trains the model on the local dataset D_i as follows:

$$g_i^{t+1} = \beta \left(\nabla F_i(D_i, w_i^{t+1}) - \frac{1}{2} \nabla F_i(w^t, D_i)^2 \right) \tag{8}$$

where g_i is the model gradient generated by the client after training the model locally. To improve the convergence speed, edge client E_i uploads gradient updates after multiple pieces of training on the local dataset.

Before uploading the gradient g_i, the i-th edge client encodeds g_i with additional Gaussian noise η_i, as shown below:

$$\hat{g}_i = g_i + \eta_i, \tag{9}$$

where η_i is Gaussian noise, $\hat{g}_i$ is the perturbed gradient, $\eta_i \sim N(0, \sigma^2)$. We have:

$$\sigma = cS_i/\epsilon_E \tag{10}$$

where ϵ_E is the privacy budget allocated by the client in a single training, c is a constant, and $c^2 > 2ln(1.25/\sigma)$. To standardize sensitivity across all clients, we apply gradient clipping to ensure that $||g_i||_2 \leq C$. C is a clipping threshold for bounding g_i. S_i is the sensitivity, which can be calculated as follows:

$$S_i = \max_{D_i} ||g_i - g_i'||_2 \leq 2C, \tag{11}$$

where D_i is the dataset of the i-th client, and g_i and g_i' are any two gradients obtained by the client's training. According to the post-processing property of differential privacy [6], RCFL satisfies differential privacy. However, in federated learning, edge clients participate in training multiple times, and the risk of client privacy leakage increases with the times of training. Therefore, it is necessary to ensure that the risk of privacy leakage of edge clients is controllable throughout the process.

We chose to use a uniform learning rate across all clients for simplicity and consistency. The decision to keep the learning rate constant across all iterations was made to avoid complicating the model updates and to maintain stability during aggregation. Since the data distribution can vary significantly across clients, especially in Non-IID scenarios, keeping the learning rate constant allows us to focus on the primary privacy-preserving mechanisms and their impact on federated learning performance. The Gaussian noise addition step plays a critical role in ensuring privacy through DP. The noise level, defined as $\sigma = cS_i/\epsilon_E$, is calculated based on the client's sensitivity S_i and the privacy budget ϵ_E. This choice of noise ensures that each gradient update is sufficiently perturbed to prevent leakage of sensitive data. While this noise improves privacy, it also introduces some trade-offs in model accuracy and convergence. Larger noise values might reduce privacy leakage, but they could slow down convergence or reduce the quality of the global model. We carefully balance these effects to ensure that privacy preservation does not excessively hinder model learning. We apply gradient

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clipping to limit the magnitude of individual gradients and prevent potential instability caused by large or anomalous updates. This is especially important in federated learning, where clients have heterogeneous data distributions and the risk of poisoned gradients or outliers is higher. By clipping gradients to a fixed threshold, we ensure that the updates remain within a stable range, promoting better convergence and robustness of the global model. The privacy budget allocation strategy is designed to control the cumulative privacy loss over multiple training iterations. Each client allocates a portion of its total privacy budget at each iteration, ensuring that the total privacy loss remains within an acceptable bound. We use different allocation methods, such as geometric and p-series, to distribute the privacy budget more efficiently, especially in scenarios where clients participate in only a subset of the total training rounds. These methods help reduce the noise added to the gradients, making the model more effective without compromising privacy.

Global Privacy Protection: RCFL ensures the global privacy of clients by controlling the privacy budget of each client throughout the training process (multiple gradient interactions). We assume that the total privacy budget for each client is ϵ, and the privacy budget allocation process can be represented as:

$$\epsilon = \sum_{t} \epsilon_t, \tag{12}$$

where ϵ_t is the privacy budget allocated by the edge client during the t-th participation in training. In traditional federated learning methods based on differential privacy, the allocation of the privacy budget is uniformly distributed according to the total number of global model iterations T. However, in federated learning for edge computing, edge clients may not participate in all training processes due to bandwidth and network fluctuations, and directly distributing the budget uniformly according to the number of model iterations would introduce a large amount of noise, thereby greatly reducing the utility of the model. Therefore, a privacy budget allocation for edge clients needs to be investigated based on the characteristics of the times of training for edge clients.

Multiple Privacy Budget Allocation: Based on the above analysis, our original problem can be transformed into how to allocate the privacy budget under a given total privacy budget to obtain the global optimal solution:

$$w^{t+1} = \sum_{i=0}^{n} g_i^t + \eta_i^t, \tag{13}$$

where η_i is the noise added to the gradient by the i-th edge client to satisfy ϵ_t-DP, thereby making it satisfy $\epsilon = \sum_t \epsilon_t$-DP throughout the process.

The privacy budget allocation problem is a series summation problem, so it can be set according to common series (such as geometric series, and p-series). The privacy budget allocated according to a geometric series is as follows:

$$\epsilon = \sum_{t=1} \epsilon_t = \sum_{t=1} q^t, \tag{14}$$

where q^t is the privacy budget allocated by the client for the t-th participation in training.

The privacy budget allocated according to a P-series is as follows:

$$\epsilon = \sum_{t=1} \epsilon_t = \sum_{t=1} \frac{1}{t^p}, \tag{15}$$

where $p >$ is a fixed constant, and $\frac{1}{t^p}$ is the privacy budget allocated by the client for the t-th participation in training. In the next section, we will specifically demonstrate the security of different allocation methods and evaluate their utility.

If client E_i participates in every training in FL, then the privacy budget equalization $\frac{\epsilon}{T}$ adds the smallest amount of noise. However, in edge computing, edge clients only participate in part of the training due to communication overhead and network fluctuation issues. Therefore, the actual number of participations ξ is uncertain and may be far less than the total number of times ($\xi << T$). Therefore, by flexibly allocating the privacy budget according to the client's online expectations and distribution, the noise added will be smaller.

Privacy Encoding: DP mechanism can perturb the gradients and reverse the flag of each element with a certain possibility, however, to trade off utility and privacy, most elements of gradient still contain abundant private features. From the results of multiple privacy leakage attacks on FL frameworks based on differential privacy, there is a compromise between utility and security [35]. Attackers may infer the belonging category, especially as the number of data interactions increases, and the risk of privacy leakage significantly increases (the attack success rate exceeds 99.5% after 20 iterations). Therefore, it is necessary to further mask the gradient to prevent privacy leakage. We propose a privacy encoding mechanism to mask the gradient information uploaded by edge clients. The specific process of the privacy encoding mechanism is as follows:

The i-th edge client locally encodes the noise-added gradient $\hat{g}_i$ according to the following rules:

$$\hat{g}_i^* = ENC(\hat{g}_i). \tag{16}$$

Specifically, the process for each element is

$$\hat{g}_{i,j}^* = \begin{cases} -1, & \text{if } \hat{g}_{i,j} \leq 0 \\ 1, & \text{if } \hat{g}_{i,j} \geq 0 \end{cases} \tag{17}$$

where $\hat{g}_{i,j}$ is the j-th element of the gradient $\hat{g}_i$, $\hat{g}_i^*$ is the leveled gradient.

The privacy encoding mechanism effectively eliminates the private features of edge clients and the impact of non-independent and identically distributed datasets contained in the gradient by leveling the gradient. Moreover, the encoded gradient does not require floating-point types but only Boolean types, thereby greatly reducing the communication overhead of edge clients.

Global Privacy Aggregation: Not only do the gradients uploaded by the client expose privacy, but the adversary can also obtain the private information of a specific client from the aggregation model through a model inversion attack, so we also need to ensure the privacy information in the aggregation model [35], [42]. Therefore, after aggregating the gradients uploaded by all edge clients, the central server determines whether additional noise needs to be added based on the privacy

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budget allocation of the clients, and broadcasts the aggregated parameters w^t to all online edge clients.

$$w^t = \sum_{i=0}^{n} \tau \hat{g}_i^{*,t-1} + \eta^t \tag{18}$$

where $\hat{g}_i^{*,t-1}$ is the leveled gradient of the client in this round, $\tau = \frac{1}{(t+1)n\epsilon}$ is the learning rate, n is the number of clients, and η^t is the noise added by the central server in this round. According to the post-processing property of DP [6], RCFL can prevent client privacy leakage in multiple information interactions. Additionally, through privacy encoding and global privacy aggregation, edge clients can greatly reduce communication overhead, thus achieving communication-efficient federated aggregation. Global privacy aggregation also ensures the prevention of privacy leakage caused by global overfitting.

V. Analysis of RCFL

This section analyzes the security, convergence, and communication overhead of RCFL.

A. Global Privacy Protection

We protect model parameter privacy during uploads and broadcasts through multiple data interactions through multiple privacy budget allocations and privacy encoding mechanisms.

In differential privacy, the scale of the noise is directly related to the allocation of the privacy budget. Therefore, we protect global privacy and obtain better global model robustness through optimized privacy budget allocation. Theorem 1 proves that RCFL ensures the privacy of a single interaction of the targeted client, Theorem 2 proves that RCFL ensures the privacy during multiple submissions of the client, and Theorem 3 proves that RCFL ensures the privacy of information in the broadcast channel. Thus, we demonstrate that RCFL can effectively protect global privacy.

Theorem 1: In RCFL, each local online client E_i's uploaded $\hat{g}_i^*$ satisfies (ϵ_t, δ_E)-DP.

Proof: To prove that the $\hat{g}_i^*$ uploaded by E_i in one iteration satisfies $(\epsilon_t,\ \delta_E)$-DP, the client E_i adds noise η_i to the local gradient $\hat{g}_i$. $\eta_i \sim N(0, \sigma_E^2)$, where σ_E can be expressed as

$$\sigma_E = cS_i/\epsilon_t, \tag{19}$$

Here S_i is the sensitivity of g_i, and ϵ_t is the privacy budget allocated by E_i in one iteration. We have

$$\sigma_E = \frac{cS_i}{\epsilon_t} = \frac{2cCw_i}{\epsilon_t}, \tag{20}$$

where the sensitivity $S_i = ||\max_{D_i}(g_i - g_i')||_2 = 2C$, $C = \max||g_i||_2$, w_i is the weight. When $\eta_i \sim N(0, (\frac{2cC}{\epsilon_t})^2)$ and $\delta_E = 1.25\exp(\frac{-\epsilon^2\sigma^2}{2\Delta f^2})$, w_i satisfies each iteration's $(\epsilon/K, \delta_E)$-DP. ■

Theorem 2: In RCFL, the gradients uploaded multiple times by edge clients satisfy (ϵ, δ)-Differential Privacy (DP).

Proof: In the RCFL framework, each edge client E_i enhances privacy by adding Gaussian noise $\eta_i^t \sim N(0, \sigma_E^2)$ to the gradient g_i^t. The noise scale σ_E is determined by

$$\sigma_E = \frac{cS_i}{\epsilon_t} = \frac{2cCw_i}{\epsilon_t} = \frac{2cC}{M\epsilon_t},$$

where $c = \sqrt{2ln(1.25/\delta)}$, representing a factor derived from privacy parameters, C is the threshold for gradient clipping, w_i is the weight of gradient g_i, and $g_i = \frac{1}{M}$, $M = |D_i|$ being the size of the dataset D_i. After adding noise, client E_i encodes the gradient for privacy. According to the post-processing property of differential privacy [7], the gradient uploaded by E_i in the t-th iteration satisfies (ϵ_t, σ)-DP. Hence, after T iterations of model updates, the cumulative privacy budget over multiple gradient submissions by the client adheres to $(\sum_{t=0}^{T}\epsilon_t, \sigma)$-DP. Since $\sum_{t=0}^{T}\epsilon_t = \epsilon$, it follows that the gradients uploaded multiple times by the client satisfy (ϵ, σ)-DP, thereby ensuring robust privacy protection over multiple interactions. According to Definition 4, encoded gradient $\hat{g}_{i,j}^*$ also satisfies (ϵ, σ)-DP. ■

Theorem 3: In RCFL, the gradients within the broadcast channel satisfy (ϵ, δ)-Differential Privacy.

Proof: Within the RCFL scheme, the central server aggregates gradients from edge clients as follows:

$$\begin{aligned} w^t &= \sum_{i=0}^{n} \tau \hat{g}_i^* + \eta \\ &= \sum_{i=0}^{n} \tau \hat{g}_i + \sum_{i=0}^{n} \tau \eta_i + \eta \end{aligned} \tag{21}$$

where each η_i is noise added to client gradients, following $\eta_i \sim N(0, \sigma_E^2)$, $\tau = \frac{1}{(t+1)n\epsilon}$, and $\sigma_E = \frac{2cC\tau}{M\epsilon_t}$, with $c = \sqrt{2ln(1.25/\delta)}$, $w_i = \frac{1}{M}$, $M = |D_i|$ representing the dataset size.

First, we calculate the global sensitivity as

$$S = \max_{D}\left\{\frac{S_i}{n}\right\} = \frac{2C\tau}{Mn}, \tag{22}$$

where $\frac{C}{M} = \max||g_i||$, C being the gradient clipping threshold, n the total number of clients. Since $\frac{2cC\tau}{M\sqrt{n\epsilon}} > \frac{2cC\tau}{Mn\epsilon}$, the CS doesn't need to add additional noise η. The gradients in the broadcast channel in RCFL satisfy (ϵ, δ)-DP. This ensures that the aggregated information remains private, even when broadcasted to all clients, thereby upholding privacy across the network. ■

B. Convergence Analysis of RCFL

For a thorough understanding of RCFL's convergence, we base our analysis on the following assumptions about the model loss function.

Assumption 1: We consider the global loss function $GF(\bullet)$, defined as $GF(\bullet) = \sum_{i=1}^{N} p_i ENC(\nabla F_i(\bullet))$, and the local loss function $F_i(\bullet)$ of each client E_i [35]:

1) $F_i(w)$ is convex.
2) $F_i(w)$ satisfies the Polyak Lojasiewicz condition with the positive parameter l;
3) $GF(w) - GF(w^*) \leq \frac{1}{2^{-l}}||w^* - w||^2$, where w^* is the optimal solution;

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4) $GF(w) - GF(w^*) = \Theta$;
5) F_i is β-Lipschitz and p-Lipschitz smooth.
6) For any i and w, $||\nabla F_i(w) - \nabla GF(w)|| \leq \varepsilon_i$, where ε_i is the divergence metric.

In RCFL, CS receives the perturbed parameters from all online clients. According to Assumption. 1 and Section III-A, we demonstrate the convergence of the global loss function as shown in Theorem 4.

Theorem 4: The global loss function converges to $\hat{w}$, and we have

$$\lim_{k\to\infty} GF\left(w^{(k)}\right) = GF(\hat{w}),$$

where $\hat{w}$ is the optimal result, and $w^{(k)}$ is the parameter vector at $k-$th iterations.

Proof: The standard gradient descent update is modified due to the encoding of the gradient. For each component j of the parameter vector of client i at iteration t, the update rule is:

$$w_{i,j}^{t+1} = w_{i,j}^{t} - \alpha \cdot \text{ENC}(\nabla F_{i,j}(w^t))$$

The quantization error for each gradient component is given by:

$$e_{i,j}^{t} = \nabla F_{i,j}(w^t) - \text{ENC}(\nabla F_{i,j}(w^t))$$

This represents the difference between the actual gradient and the encoded version.

Using the Lipschitz continuity of F_i, the error can be bounded. If F_i is β-Lipschitz, then for each component j:

$$|e_{i,j}^{t}| \leq |\nabla F_{i,j}(w^t)| \leq \beta$$

First, express $GF(w^{t+1})$ in terms of $GF(w^t)$ and the gradient updates. Given the quantized update rule, we have:

$$GF(w^{t+1}) = GF(w^t - \alpha \cdot \text{ENC}(\nabla F(w^t)))$$

Rewrite this expression to include the quantization error. Remember that the actual gradient descent step would have been $w^t - \alpha\nabla F(w^t)$, but due to quantization, it is $w^t - \alpha \cdot \text{ENC}(\nabla F(w^t))$.

If GF is Lipschitz continuous with constant L, you can bound the change in GF due to a change in w. This leads to:

$$|GF(w^{t+1}) - GF(w^t)| \leq L\|w^{t+1} - w^t\|$$

Substitute the quantized gradient update rule into the inequality:

$$|GF(w^{t+1}) - GF(w^t)| \leq L\alpha\|\text{ENC}(\nabla F(w^t))\|$$

This condition helps relate the decrease in $GF(w)$ to the distance from the optimal solution. Mathematically, it can be shown that:

$$GF(w^{t+1}) - GF(w^*) \leq (1 - 2\alpha l) \cdot (GF(w^t) - GF(w^*))$$

where l is the Polyak-Lojasiewicz parameter, α is the learning rate, and w^* is the optimal solution.

Utilizing the bounds and conditions established in previous steps, we can show that:

$$\lim_{t\to\infty} GF(w^t) = GF(w^*)$$

Combine all the above elements to conclude that as t increases, w^t converges to w^*, and consequently, $GF(w^t)$ converges to $GF(w^*)$. ■

Although the encoded gradient may not make the global model learn in the direction of the steepest descent, this encoding mechanism will also weaken the impact of non-IID and overfitting on model convergence. Therefore, through certain adjustments, RCFL can achieve better convergence effects in federated learning for edge computing.

C. Communication Overhead and Computational Overhead

In this section, we provide a detailed complexity analysis of RCFL framework, focusing on both computational complexity and communication complexity. This analysis highlights the efficiency of RCFL, especially in resource-constrained edge computing environments. The computational complexity of RCFL is primarily driven by two main components: the local training performed by edge clients and the aggregation performed by the central server. Each edge client performs local model training on its private dataset. The computational cost for this step is proportional to the dataset size D_i, the model size M, and the number of training iterations T. The complexity of each client's local training is approximately $\mathcal{O}(D_i \cdot M \cdot T)$, Where D_i is the number of training samples at client i, M is the number of model parameters, T is the number of local training iterations. The central server aggregates the gradients received from all clients and broadcasts the updated model. The aggregation complexity is $\mathcal{O}(n \cdot M)$, which is linear with respect to the number of clients n and the model size M. The total computational complexity for RCFL per iteration is dominated by the local training of all clients and the aggregation at the central server. The overall complexity is $\mathcal{O}(n \cdot D_i \cdot M \cdot T) + \mathcal{O}(n \cdot M)$. Communication complexity in RCFL is driven by the amount of data exchanged between the clients and the central server. This includes the transmission of gradient updates from the clients to the server and the broadcast of the global model from the server to all clients. Each client sends its perturbed gradient to the central server. The size of the gradient update depends on the model size M and the encoding mechanism. Since each gradient element is encoded into a single bit, the communication cost per client is reduced to $\mathcal{O}(M)$. For n clients, the total communication cost for transmitting gradients is $\mathcal{O}(n \cdot M)$. After aggregating the gradients, the central server broadcasts the updated model to all clients. The communication cost for broadcasting the model is proportional to the model size M and the number of clients n. The total communication complexity per iteration is $\mathcal{O}(n \cdot M)$. The complexity of gradient clipping is constant per client, as it involves comparing the gradient magnitude with a predefined threshold. The complexity is $\mathcal{O}(M)$. The complexity of adding Gaussian noise is linear with respect to the number of gradient elements, as each element must receive an independent noise sample. The complexity is $\mathcal{O}(M)$. The privacy encoding mechanism reduces the communication size by converting gradient values into binary form, which requires $\mathcal{O}(M)$ operations per client. The overall computational complexity per iteration is:

$$\mathcal{O}\left(n \cdot D_i \cdot M \cdot T\right) + \mathcal{O}\left(n \cdot M\right) \tag{23}$$

The overall communication complexity per iteration is:

$$\mathcal{O}\left(n \cdot M\right) \tag{24}$$

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This analysis shows that RCFL is computationally efficient, particularly in edge computing environments where the communication and computational resources are limited. The key advantage of RCFL is its ability to significantly reduce the communication overhead via the privacy encoding mechanism while ensuring privacy protection through differential privacy techniques. RCFL significantly accelerates the convergence speed, reduces the communication overhead in FL, and is beneficial for edge clients with limited resources. This efficiency is achieved through multiple privacy budget allocations, combined with a novel privacy encoding technique. RCFL's approach to communication efficiency makes it particularly suitable for deployment in distributed machine-learning scenarios involving edge devices. The communication overhead of edge clients in RCFL is observed to be less than one-tenth of that required by traditional methods like MID [32], BiDO [26], and class-DP [40]. RCFL introduces a privacy encoding mechanism that reduces the communication overhead to approximately $O(\frac{d}{10})$ of the original requirement, where d represents the dimension of the gradients. In RCFL, the adoption of a privacy encoding mechanism by edge clients is pivotal. This encoding strategy effectively allows each gradient element to be represented using just one bit (-1 or 1), as opposed to the traditional multi-bit representation. Traditionally, gradients are represented with floating-point numbers. The encoding reduces this requirement to a single bit per gradient element. This significant reduction in data transmission requirements makes RCFL highly efficient for edge computing scenarios, where bandwidth and computational resources are often limited. The reduced communication overhead accelerates the learning process and conserves bandwidth and energy resources, which are important in edge computing environments. This efficiency enables more frequent model updates and improved real-time learning capabilities without overburdening the network or edge devices.

In conclusion, through the privacy encoding mechanism, RCFL achieves a remarkable reduction in communication overhead, making FL more practical and efficient in resource-constrained edge computing environments.

VI. Evaluation

This section provides a comprehensive evaluation of our proposed RCFL compared to state-of-the-art privacy-preserving FL schemes such as MID [32], BiDO [26], and Class-DP [40]. These schemes are particularly relevant in defending against privacy leakage attacks. We conduct a series of experiments on two image datasets, focusing on aspects of security, convergence, and communication overhead for edge computing.

A. Experimental Setup

We conducted our experiments on two well-known datasets: MNIST[3] and CIFAR-10,[4] chosen for their widespread use in evaluating machine learning models. The details of the experimental setup, including the datasets, model network, and equipment environment, are presented in Tables II, III, and IV respectively.

[3]http://yann.lecun.com/exdb/mnist/

[4]https://www.cs.utoronto.ca/kriz/cifar.html

TABLE II
Datasets Used in Experiments

Dataset	MNIST	CIFAR-10
Type	Grayscale Images	Color Images
Training Samples	60,000	50,000
Test Samples	10,000	10,000
Training Samples Per Client	1-128	1-2,000
Test Samples Per Client	100	100

TABLE III
Model Network Specifications

Model	Multilayer Perceptron (MLP) Network
Per Hidden Layer	256 Units
Activation Functions	ReLU
Learning Rate	0.002
Network Depth	3-18 Layers

TABLE IV
Equipment and Environment Specifications

Equipment	High-Performance Desktop
Operating System	Ubuntu 16.09 LTS (64-bit)
CPU	Intel i5-13400
GPU	NVIDIA 3090
RAM	32 GB

TABLE V
Privacy Leakage Indicated by Attack Accuracy and Attack Success Rate Over T Iterations on MNIST Against MPLA (ASR 10 Means Attack Success Rate of MPLA After 10 Iterations)

Scheme	Accuracy	ASR 10	ASR 20
MID	80.65	98.31	99.83
BiDO	86.67	82.12	88.56
Class-DP	91.80	96.15	99.23
RCFL	93.58	41.12	49.63

TABLE VI
Privacy Leakage Indicated by Attack Accuracy and Attack Success Rate Over T Iterations Against MPLA on CIFAR 10 (ASR 10 Means Attack Success Rate of MPLA After 10 Iterations)

Scheme	Accuracy	ASR 10	ASR 20
MID	76.81	99.78	100.00
BiDO	87.39	99.35	99.97
Class-DP	90.61	99.92	100.00
RCFL	92.31	86.18	94.85

B. Multiple Privacy Leakage Attack Robustness

Tables V and VI show how different schemes perform against MPLA on the MNIST and CIFAR-10 datasets. These tables, along with Fig. 4, illustrate the success rates of MPLA under various conditions. We set parameters $\epsilon = 0.1$, $\aleph = 0.1$, and $n = 500$ to measure the MPLA success rates. The results highlight RCFL's superior performance, especially when using clipping (privacy encoding). This approach significantly lowers attack success rates, proving RCFL's effectiveness in MPLA defense. Other methods like Class-DP, MID, and BiDO show vulnerabilities under MPLA. Class-DP focuses on adding noise to gradients and decoupling inputs from outputs, but they face increasing privacy risks over multiple training iterations. BiDO improves upon MID's model but does not fully counter gradient issues in non-IID data scenarios. RCFL adopts a dual strategy of

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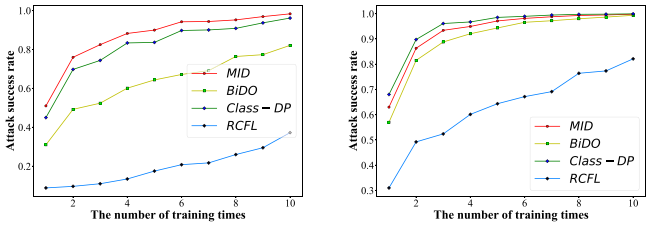


Fig. 4. (a) shows the success rate of multiple privacy leakage attacks with various training times on dataset MNIST; (b) shows the success rate of multiple privacy leakage attacks with various training times on dataset CIFAR.

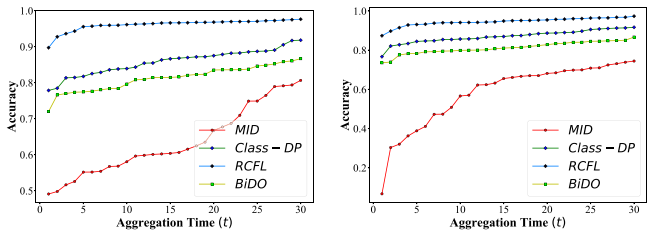


Fig. 5. (a) shows the accuracy for each scheme with various aggregation times on dataset MNIST; (b) shows the accuracy for each scheme with various aggregation times on dataset CIFAR.

optimized privacy budget allocation and innovative privacy encoding. This combination effectively anonymizes gradient information, providing protection against MPLA even with repeated training. Our evaluations show that RCFL beats other methods like MID, BiDO, and Class-DP by significantly reducing the MPLA success rate.

In conclusion, RCFL stands out in reducing privacy risks associated with both single and multiple data releases, particularly in non-IID data scenarios within FL environments. Our findings confirm RCFL's robustness in protecting FL models from sophisticated privacy attacks, making it an ideal choice for demanding privacy preservation scenarios.

C. Evaluation of Convergence

Fig. 5 and Table V show our evaluation of convergence using average accuracy from five experiments on MNIST and CIFAR-10 with $\epsilon = 0.05$. This assessment reveals significant insights into the performance and privacy trade-offs of various FL schemes. MID aims to protect client data by reducing the input-output correlation, which serves as a defense against privacy attacks [32]. However, this reduction compromises the model's classification accuracy, leading to diminished performance compared to less restrictive FL methods. BiDO retains more of the model's predictive capabilities by obfuscating the input-output relationship, improving performance over MID but still impacting overall accuracy due to the introduced ambiguity [26]. Class-DP alters its classification process to preserve data features while minimizing misclassification risks, thus maintaining better predictive performance [40].

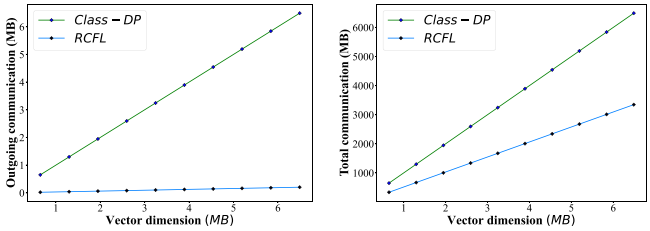


Fig. 6. (a) shows the outgoing communication of per edge client at one iteration; (b) shows the total communication of the central server at one iteration.

Our RCFL without clipping method addresses both privacy protection and convergence efficiency. However, random DP noise may introduce greater noise and further affect the convergence of the model. RCFL without DP uses the sign of gradients to update the global model, which reduces the effect of overfitting and non-IID datasets. However, the adversary who knows the direction of gradient descent can also use MPLA to infer the private information of edge clients. RCFL with clipping and DP combines enhanced privacy budget allocation with advanced privacy encoding to perturb the direction of gradient descent, thereby reducing the effect of non-IID datasets and overfitting, resulting in strengthened data privacy. Although adding noise reduces the convergence of RCFL, the ASR reduces decreases more than 2%. Experimental results on MNIST and CIFAR demonstrate that RCFL outperforms MID, BiDO, and Class-DP. This significant enhancement in both privacy and effectiveness makes RCFL a robust choice for secure and efficient FL.

D. Evaluation of Communication Overhead

As demonstrated in Fig. 6, RCFL significantly transforms FL in edge computing through its novel privacy encoding mechanism. This technique compresses communication data to approximately 1/32 of the original size, enabling transformative network resource utilization, especially critical in bandwidth-limited edge environments with power constraints. The encoding process simplifies gradient data transmission during learning by converting full precision gradients into a binary format. Each gradient is represented as a single bit, indicating its direction, thus considerably reducing data size.

Comparatively, methods like MID [32], Class-DP [40], and BiDO [26] exhibit similar communication overheads due to transmitting full or significant portions of gradients. In contrast, RCFL's bandwidth requirements are substantially lower, resulting in less than a tenth of the communication load of these methods. Beyond data reduction, RCFL's privacy encoding also enhances data privacy by minimizing information exchange and adding an obfuscation layer, which mitigates the risk of sensitive data exposure and inference attacks. The reduced communication overhead contributes to faster training times and lower energy consumption, aligning with the core objectives of edge computing: efficiency and scalability.

In summary, RCFL excels in FL by efficiently balancing communication and privacy preservation. Its combination of privacy

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encoding and optimized algorithms establishes a new standard in FL, particularly for resource-constrained edge environments.

VII. Related Work

This section outlines preceding methodologies in FL for edge computing and privacy-preserving FL, with a particular emphasis on mitigating data leakage.

A. FL for Edge Computing

Conventional centralized machine learning models rely on third-party entities to aggregate data and facilitate model training, subsequently providing services. This centralized data exchange paradigm introduces latency issues and compromises service quality due to the inherent delay in cloud communication. Edge computing, characterized by its distributed architecture, addresses these concerns by processing data at the network's periphery, ensuring expedited communication responses. The Internet of Things devices, in tandem with nearby edge servers, leverage these servers' computational resources, thereby enhancing communication reliability and computational efficiency. Despite these advancements, the deployment of extensive model parameters, potentially in the trillions, to IoT devices presents a formidable challenge. Moreover, the integrity of edge client data is at risk during transmission, leaving it vulnerable to a spectrum of adversarial attacks [39], [2]. To fortify data privacy within edge computing, FL compels IoT devices to conduct model training locally, sharing intermediate model updates with edge servers, thus eschewing the transmission of raw data. Termed FL for edge computing [11], this methodology amalgamates edge computing with FL, enabling model training proximal to data sources, bolstering communication efficiency, and safeguarding data privacy. However, the encrypted transmission of large global model parameters remains a significant challenge for resource-limited edge clients. Consequently, there is a pressing need to devise a robust FL framework for edge computing that guarantees security, efficiency, and operational efficiency.

B. Privacy-Preserving FL

Addressing privacy concerns, a common tactic involves adding artificial noise to data, a method illustrated by DP strategies [7]. Research in DP-focused machine learning algorithms includes Local Differential Privacy (LDP) [8], DP-optimized Stochastic Gradient Descent (SGD) [1], DP-based parameter update methodologies [35], and DP-adherent meta-learning [17].

LDP requires clients to locally alter model information, sending only modified parameters or gradients instead of raw data [30]. Another approach involves estimating client data uploads using LDP to protect this data [31]. For large-scale neural networks, DP-focused training algorithms have been developed, like the one in [23] for improved word prediction in complex scenarios. The cumulative privacy loss in models is calculated by aggregating individual privacy losses, with [1] improving computational efficiency in this regard. An innovative DP-based SGD algorithm that establishes performance benchmarks relative to privacy levels and data sizes is presented in [36]. Furthermore, [17] introduces an SGD meta-learning algorithm aligning with DP, showcasing a convex optimization landscape. The tensor-based chain abstraction model from [28] effectively obscures complex errors while maintaining model data privacy through DP and multi-party computation. However, the noise scale added by the above methods cannot affect the opponent's classification of target data labels within a certain threshold, and excessively large-scale noise significantly affects model utility. [40] adds differential privacy noise to model outputs at various levels and refers to different labels with a certain possibility while involving less noise.

Another way to protect privacy is to add a penalty function to the model to reduce the correlation between the model output and the input [32]. Yet, this reduction can adversely affect model utility. When the correlation drops below a certain level, the model fails to learn from inputs. Calculating mutual information also demands extensive computational resources, challenging for resource-constrained devices. To balance input-output correlation reduction and model utility, [26] integrates dependency constraints in training, enhancing accuracy and defending against privacy attacks. Another approach,

Standard solutions cater to centralized or single-release privacy, but edge clients often contain sequentially correlated data (e.g., series of personal photos), making participant privacy more vulnerable. Thus, methods effective for single data releases might not suffice against multiple privacy leakage. We investigate multiple privacy leakage attacks targeting FL in edge computing. We reveal the ineffectiveness of existing defenses against MPLA, with a single model inversion attack success rate of 31%, escalating to 82.12% after ten releases, and 88.56% after twenty. To counter MPLA, we propose a robust, communication-efficient FL framework (RCFL) for edge computing, featuring a global privacy-preserving mechanism and privacy encoding to safeguard privacy during continuous data releases and ensure communication efficiency.

VIII. Conclusion

In this paper, we presented RCFL, a robust and communication-efficient FL framework designed for edge computing environments to mitigate MPLA. Unlike traditional FL approaches, which are vulnerable to repeated privacy attacks, RCFL employs a novel privacy encoding strategy combined with a global privacy-preserving mechanism. This ensures differential privacy throughout multiple data releases while substantially reducing communication overhead for resource-constrained edge devices. Our comprehensive theoretical analysis demonstrates that RCFL maintains differential privacy across multiple interactions without compromising model convergence. Experimental results on the MNIST and CIFAR-10 datasets further validate the effectiveness of RCFL. Compared to state-of-the-art methods such as MID, BiDO, and Class-DP, RCFL reduces the MPLA success rate from 88.56% to 42.57%. Additionally, RCFL achieves over a 90% reduction in communication costs, highlighting its practical advantages in enhancing security, efficiency, and scalability for FL in edge computing scenarios.

Future work will investigate adaptive privacy mechanisms to dynamically balance the trade-off between privacy preservation

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and model utility, further optimize the privacy encoding strategy, and extend the approach to other machine learning tasks beyond image classification. These advancements could enhance the robustness of FL systems against evolving privacy threats while maintaining high model performance in diverse and resource-constrained environments.

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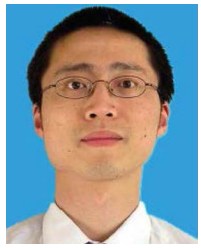
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